

Intak Hwang

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GitHub: [sp301415](https://github.com/sp301415)

I'm a Ph.D. student in Cryptography & Privacy Lab at Seoul National University, advised by Yongsoo Song. I'm interested in post-quantum cryptographic protocols based on lattices, including but not limited to Fully Homomorphic Encryption and Zero-Knowledge Proofs.

EDUCATION

Seoul National University 2023 — Present
Integrated M.S./Ph.D. in Computer Science & Engineering
Advised by Yongsoo Song

DGIST 2018 — 2022
B.S. in School of Undergraduate Studies *Summa Cum Laude*

PUBLICATIONS

- [C09] 2025/1804
HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping
Binwu Xiang, Seonhong Min, Intak Hwang, Zhiwei Wang, Haoqi He, Yuanju Wei, Kang Yang, Jiang Zhang, Yi Deng, Yu Yu
Eurocrypt 2026
- [C08] 2024/2032
Carousel: Fully Homomorphic Encryption with Bootstrapping over Automorphism Group
Intak Hwang, Seonhong Min, Yongsoo Song
Asiacrypt 2025
- [C07] 2025/382
On the Security and Privacy of CKKS-based Homomorphic Evaluation Protocols
Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song
Asiacrypt 2025
- [C06] 2025/216
Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation
Intak Hwang, Jinyeong Seo, Seonhong Min, Yongsoo Song
ACM CCS 2025
- [C05] 2024/1879
Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption
Intak Hwang, Hyeonbum Lee, Jinyeong Seo, Yongsoo Song
ACM CCS 2025
- [C04] 2025/1255

Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition

Intak Hwang, Shinwon Lee, Seonhong Min, Yongsoo Song
SAC 2025

[C03] 2024/1502

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

Hyunho Cha, Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song
IEEE S&P 2025

[C02] 2024/306

Concretely Efficient Lattice-based Polynomial Commitment from Standard Assumptions

Intak Hwang, Jinyeong Seo, Yongsoo Song
Crypto 2024

[C01] 2023/1328

Optimizing HE via Level-aware Key-switching

Intak Hwang, Jinyeong Seo, Yongsoo Song
WAHC 2023

[J01] **A Privacy-Preserving HLA Imputation Method with Homomorphic Encryption**

Hakin Kim, Intak Hwang, Yongsoo Song, Buhm Han
iScience

PREPRINTS

[P03] 2026/044

Jindo: Practical Lattice-Based Polynomial Commitments for Client-Side Proving

Intak Hwang, Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

[P02] 2025/395

Provably Secure Approximate Computation Protocols from CKKS

Intak Hwang, Yisol Hwang, Miran Kim, Dongwon Lee, Yongsoo Song

[P01] 2025/203

Ciphertext-Simulatable HE from BFV with Randomized Evaluation

Intak Hwang, Seonhong Min, Yongsoo Song

PROJECTS

TFHE-go (GitHub Repository)

TFHE-go is an implementation of (MK)TFHE scheme, written in Go and Go Assembly. Currently, it is one of the fastest and most feature-complete TFHE implementaion available open-source.

Ringo-SNARK (GitHub Repository)

Ringo-SNARK is a Zero-Knowledge PIOP toolkit for efficiently proving Ring-LWE relations, written in Go. It supports simple, gnark-like circuit design and compilation.

PRESENTATIONS

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption
FHE.org 2026 — Taipei, Taiwan

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation
ACM CCS 2025 — Taipei, Taiwan
FHE.org — Online

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing
IEEE S&P 2025 — San Fransisco, USA

Optimizing HE via Level-aware Key-switching
WAHC 2023 — Copenhagen, Denmark

HONORS AND SCHOLARSHIPS

National Cryptographic Contest	
Excellence Award, Encouragement Award	2025
Best Award, Excellence Award	2024
Special Award	2023

CTF Security Competitions	2020 — 2022
SSTF Hacker's Playground 2022	5th place
WhiteHat Contest 2021	3rd place
DEFCON CTF 2021	Finalist
PlaidCTF 2021	5th place
Real World CTF 2020/2021 (Media Coverage)	1st place
Midnight Sun CTF 2020 Finals	7th place
TokyoWesterns CTF 2020 Finals	3rd place
DEFCON CTF 2020	Finalist

DGIST Dean's List	2020
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SKILLS

Languages

Korean (native), English (fluent)

Programming Languages

Go, Python (SageMath), C/C++, C#, Rust, $\text{\LaTeX}$

OTHER ACTIVITIES

Member of CTF Team CodeRed

2020 — 2022

I participated in CTF competitions from time to time, mostly solving crypto challenges.

Developer & Writer of Team Invertible

2020 — Present

I am actively working on *Shards of Time*, a sokoban puzzle game. We are planning to release the game on Steam.

OTHER INTERESTS

I love watching films. I've directed several short films, and I'm eager to continue doing so.